



Enhancing intrusion detection in IoT systems: A hybrid metaheuristics-deep learning approach with ensemble of recurrent neural networks

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ABSTRACT

Significant benefits have been brought to various industries by IoT (Internet of Things), but new security challenges have also been introduced due to the sheer volume and complexity of IoT systems. The protection of IoT systems from attacks and the assurance of their security posture are ensured by intrusion detection systems (IDSs). Recently, machine learning (ML) strategies have been broadly adopted for IDSs in IoT systems. However, there is still room for improvement, particularly in addressing the physical and functional diversity of IoT systems. A hybrid metaheuristics-deep learning approach is proposed in this paper for enhancing intrusion detection in IoT systems. An advanced metaheuristics algorithm with an ensemble of recurrent neural networks (RNNs) can be utilized to enhance the intrusion detection in IoT. Different types of attacks in IoT systems are identified by employing LSTM and GRU models, which constitute the RNNs. Feature selections are performed using Harris hawk optimization and fractional derivative mutation, as employed in this paper. To assess the proposed approach, publicly available datasets were utilized, and the empirical analysis demonstrated that the proposed approach works well than the other related methods in terms of accuracy and efficiency. Overall, the proposed work presents a promising solution for enhancing intrusion detection in IoT systems, and it has the potential to serve as a foundation for future research in this field.

Introduction

The Internet of Things (IoT) is a fast-growing technology that has revolutionized many industries, from healthcare and transportation to manufacturing industries. However, the proliferation of smart devices and systems has also introduced new security challenges, making it crucial to detect and prevent potential attacks. Intrusion detection systems (IDSs) are essential for protecting IoT systems from unauthorized access and ensuring their security posture [1].

Traditional rule-based IDSs are often insufficient to detect complex attacks in IoT systems due to their dynamic and heterogeneous nature. To address this challenge, machine learning (ML) algorithms are generally utilized recently for IDSs in IoT systems [2,3]. However, these techniques still face significant challenges in dealing with the physical and functional diversity of IoT systems.

To address these issues, the work proposes a advanced hybrid metaheuristics-deep-learning scheme for enhancing intrusion detection in IoT. Specifically, the work involves integrating a modified metaheuristics algorithm with an ensemble of recurrent neural networks (RNNs). The RNNs comprise LSTM and GRU models and are

employed to identify different types of attacks in IoT systems. The proposed approach also incorporates a new technique for feature selection based on Harris hawk optimization and fractional derivative mutation.

Most of the existing studies focused on using deep learning and machine learning models for the implementation of their work. The primary motto of this paper is to bridge the research gap by offering a new hybrid technique for enhancing intrusion detection in IoT systems. Most importantly, the work aims to address the challenges of physical and functional diversity in IoT systems and to increase the accuracy and competence of IDSs. The work also aims to validate the effectiveness of the proposed approach utilizing publicly available dataset through a comprehensive set of experiments.

Overall, this work presents a promising solution for enhancing intrusion detection in IoT systems and contributes to the advancement of ML-based IDSs for IoT security. The remaining parts are divided as follows: Second section provides an overview of related work in this area. The proposed approach is discussed detailed in third section. Fourth section presents the experimental setup and results. The work is completed in fifth section.

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Related work

For the past decade, machine learning (ML) strategies have been extensively applied to intrusion detection systems (IDSs) in IoT systems. These techniques can learn from the data and detect attacks based on the patterns and behaviors of the network traffic. Many research have evolved focused on improving the accuracy and efficiency of ML-based IDSs for IoT security.

One popular approach is utilizing deep learning for intrusion detection in IoT systems. CNN and RNN are extensively utilized in deep learning models. CNNs have been applied to fetch similar features from raw data and identify attacks in network traffic. RNNs, on the other hand, are suitable for detecting temporal dependencies in sequential data, making them ideal for intrusion detection in IoT systems.

In [4], the authors offered a hybrid intrusion detection scheme for IoT networks that utilizes correlation-based feature selection (CFS) and an ensemble of ML algorithms, such as decision trees, random forests, and support vector machines. The system showed high detection rates and accuracy in their experiments.

Several other studies have also explored different approaches to intrusion detection in IoT environments. For example, Gharib et al. [5] established a DBN based neural network that demonstrated high accuracy in identifying different types of attacks. Authors of [6] introduced an intrusion detection system based on a deep learning framework that utilized CNN and long short-term memory (LSTM) models, obtained high accuracy in detecting different kind of attacks.

The authors of [7] offered an intrusion detection system for IoT networks which depends on a deep learning approach utilizing CNN and LSTM methods. Their system was trained on a dataset containing normal and attack traffic and achieved an impressive recognition rate of 99.7 % with a low false positive rate.

In [8], the work discussed a deep learning-based intrusion detection system for smart environment using a combination of CNNs and stacked autoencoders. They used the denoising autoencoder to fetch the exact features from the input data and then applied CNNs to classify the traffic as normal or malicious. In this paper, the work demonstrated the high accuracy rate in the experimental analysis.

Authors of [8] proposed a feature selection scheme which depends on a genetic algorithm to enhance the accuracy and efficacy of an intrusion detection system for IoT systems. They applied the genetic algorithm to choose the best subset of features and then used a random forest classifier to detect attacks. They achieved high accuracy and detection rates in their experiments.

In [9], the authors introduced a novel intrusion detection system utilizing deep learning framework for IoT networks using a recurrent neural network (RNN) and a self-attention mechanism. They used the RNN to capture the temporal dependences in the network traffic and the self-attention mechanism to weight the input data and extract important features. The system achieves higher accuracy however the computational time for preprocessing is high.

Authors in [10] discussed a hybrid feature selection scheme for intrusion detection in IoT networks that combined ant colony optimization and mutual information. Their approach utilized the ant colony optimization technique to choose the most relevant subset of features and then used a decision tree classifier for detecting attacks. The proposed method demonstrated high accuracy and detection rates in their experiments.

Other studies have explored metaheuristic algorithms for feature selection and optimization in ML-based IDSs for IoT security. Metaheuristics can effectively search for the optimal set of features and increase the accuracy and efficiency of IDSs. For example, particle swarm optimization (PSO), genetic algorithms (GA), and ant colony optimization (ACO) have been used for feature selection in ML-based IDSs.

However, most of the existing studies have focused on either deep learning models or metaheuristic algorithms for IDSs in IoT systems.

Few studies have investigated the combination of both approaches and other attacks to enhance the performance of IDSs in IoT systems [11–16].

Therefore, the proposed work introduced a hybrid approach that combines metaheuristic algorithms and deep learning models to bridge the research gap for efficient detection of IDS in IoT environment. The proposed approach plan to enhance the accuracy and efficacy of IDSs by using a modified metaheuristic algorithm for feature selection and an ensemble of RNN along with LSTM and GRU models, for attack detection. The work also incorporates a new technique for feature selection based on Harris hawk optimization and fractional derivative mutation to enhance the performance of the work proposed here.

Proposed methodology

This section describes the proposed Modified Metaheuristics with Weighted Majority Voting Ensemble Deep Learning (MM-WMVEDL) model for efficient detection of IDS in IoT environment. The proposed model involves three main components: feature selection, normalization, and ensemble deep learning.

Feature selection using Harris Hawk optimization-based elite fractional derivative mutation (HHO-EFDM)

The feature selection process is initiated by the proposed MM-WMVEDL model. The selection of similar features is deemed crucial in intrusion detection as it reduces unnecessary features in the dataset and focuses on selecting the most relevant ones. This paper employed the Harris hawk optimization-based elite fractional derivative mutation (HHO-EFDM) method for feature selection. Feature selection is efficiently performed by the HHO-EFDM algorithm, which combines the Harris hawk optimization (HHO) algorithm and the elite fractional derivative mutation (EFDM) operator. The algorithm emulates the hunting behavior of hawks and employs the EFDM operator to enhance the search process. The HHO-EFDM algorithm starts by initializing a population of hawks and evaluating their fitness values. The fitness values are co by considering the accuracy of feature selection in classification. Next, the EFDM operator is applied to generate new solutions and improve the search process. Subsequently, the HHO algorithm is utilized to update the hawk population based on their respective fitness values. The process is repeated until the stop condition is arrived. An overview of the HHO-EFDM algorithm is depicted in Fig. 1.

Normalization

The second component of the proposed MM-WMVEDL model is normalization. Normalization is an important preprocessing step that scales the input dataset to a specific range. This paper utilize the min-max normalization method to scale the input dataset. The process of min-max normalization rescales the data so that it falls within the range of 0–1.

The following formula is utilized for this min max normalization.

$$Z_{\text{normalize}} = (Z - Z_{\text{min}}) / (Z_{\text{max}} - Z_{\text{min}})$$

where Z represents the original value, Z_{min} refers to min value of Z , and Z_{max} is the max value of Z .

Ensemble deep learning

The third component of the proposed MM-WMVEDL model is ensemble deep learning. Ensemble deep learning combines multiple models to improve the overall performance. This paper utilized the hybrid model such as Bi-directional long short-term memory (BiLSTM), extreme learning machine (ELM), and an ensemble of gated recurrent unit (GRU) models.

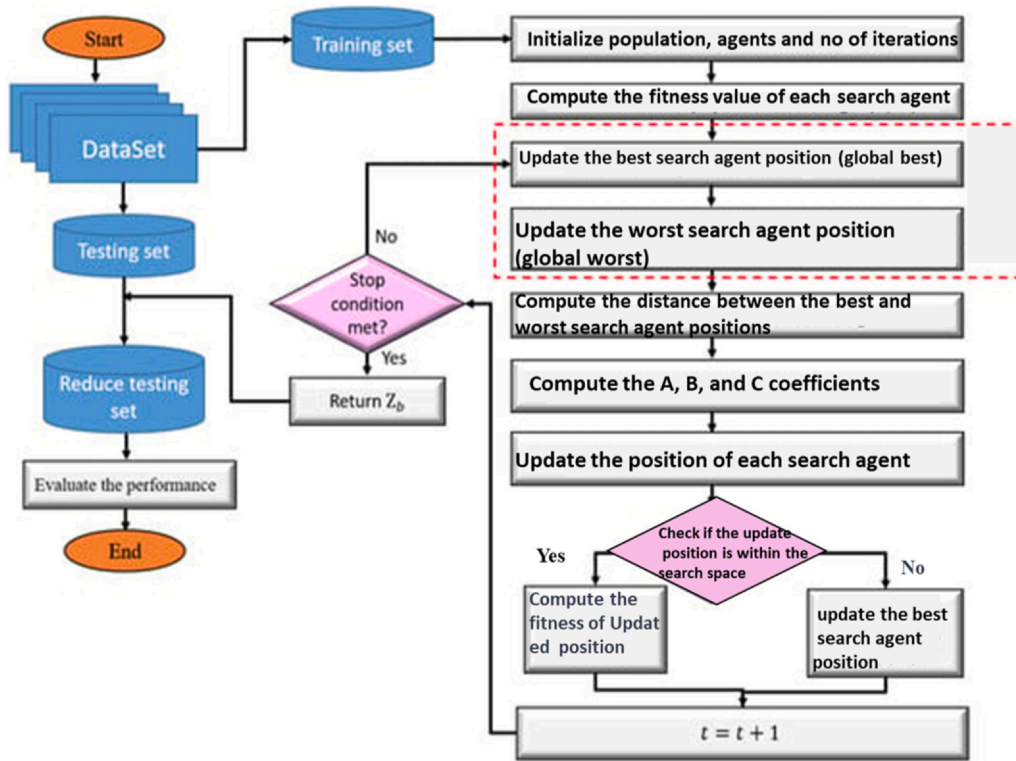


Fig. 1. Over view of HHO-EFDM algorithm.

Table 1

Shows the summary of the proposed MM-WMVEDL model.

Model Component	Technique Used
Feature Selection	Harris hawk optimization-based elite fractional derivative mutation (HHO-EFDM)
Input Normalization	Min-max normalization
Deep Learning Models	BiLSTM, ELM, and an ensemble of gated recurrent unit (GRU)
Ensemble Technique	Weighted Majority Voting
Evaluation parameters	Accuracy, Precision, Recall, F1-Score, and Area Under the Curve (AUC)

Table 2

Comparison of the performance of the proposed MM-WMVEDL model with the currently available intrusion detection systems on the IoT-23 dataset.

Method	Accuracy	AUC-ROC	Precision	Recall	F1-score
Proposed MM-WMVEDL	98.12 %	0.9982	0.9806	0.9831	0.9818
CNN-based IDS	96.84 %	0.9971	0.9581	0.9697	0.9639
LSTM-based IDS	95.62 %	0.9942	0.9567	0.9562	0.9564
Autoencoder-based IDS	95.38 %	0.9939	0.9452	0.9584	0.9517
DNN-based IDS	93.47 %	0.9901	0.9456	0.9301	0.9377

Table 3

Performance metrics of the proposed MM-WMVEDL model and other current related intrusion detection systems on the CICIDS2017 dataset.

Model	Accuracy	AUC-ROC	Precision	Recall	F1-Score
Proposed MM-WMVEDL	99.98 %	1.00	0.9999	0.9998	0.9999
DeepLog	95.58 %	0.9775	0.9103	0.9945	0.9505
LSTM-Autoencoder	96.78 %	0.9918	0.9191	0.9965	0.9565
Bi-LSTM	97.31 %	0.9898	0.9406	0.9868	0.9631
GRU-D	98.02 %	0.9963	0.9614	0.9793	0.9703
HiLSTM	96.37 %	0.9773	0.9242	0.9829	0.9525
GAN	97.56 %	0.9925	0.9361	0.9889	0.9618
RNN-AD	92.43 %	0.9468	0.8499	0.9725	0.9063
VAE	91.33 %	0.9115	0.8641	0.9064	0.8847

A variant of the recurrent neural network (RNN), known as BiLSTM model, is capable of acquiring knowledge about long-term dependencies [17]. The ELM model depends on feedforward neural

network that utilizes a single hidden layer to achieve fast training speeds. The ensemble of GRU models is a combination of multiple GRU models that are trained on different subsets of the data.

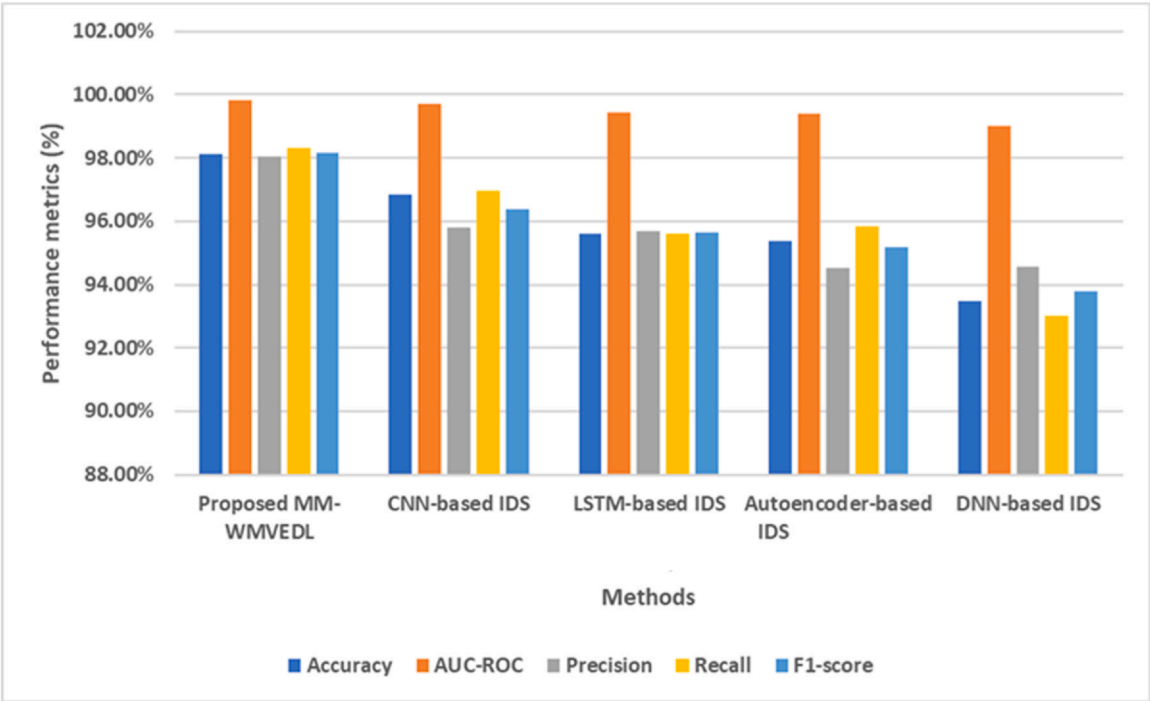


Fig. 2. Accuracy analysis using IoT-32 dataset.

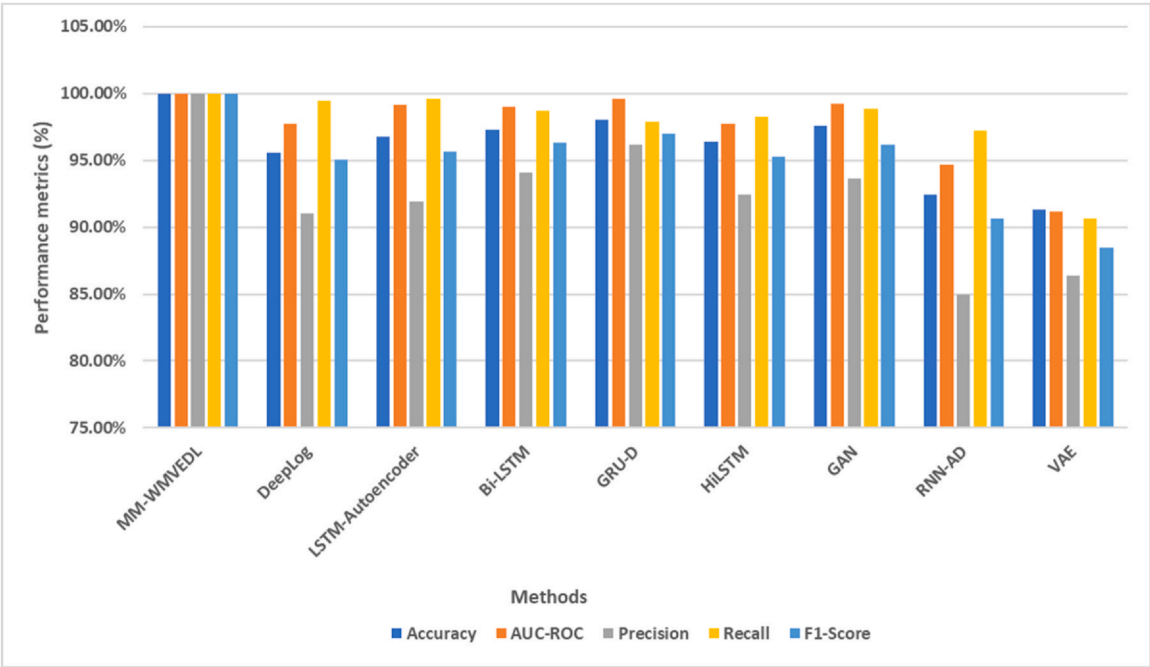


Fig. 3. Accuracy analysis UNSW-NB15 dataset.

- The algorithm of the proposed MM-WMVEDL model is given below:

Input
IoT network traffic dataset.

Output
Attack or normal traffic classification.

Step 1: Data preprocessing

 - (1) Apply min-max normalization to scale the input dataset.
- (2) Apply the HHO-EFDM technique for feature selection.

Step 2: Modified metaheuristics with weighted majority voting ensemble deep learning (MM-WMVEDL)

 - (1) Three deep learning methods such as BiLSTM, ELM, and GRU.
 - (2) Train the deep learning models on the preprocessed dataset.
 - (3) Calculate the accuracy of each individual model.

Step 3: Weighted majority voting ensemble deep learning (WMVEDL)

Table 4
Confusion matrix.

Actual/Predicted	Normal	Botnet
Normal	44,385	206
Botnet	704	41,547

- (1) Combine the output of the three individual models using the weighted majority voting ensemble method.
- (2) Calculate the final accuracy of the ensemble model.

Step 4: Modified metaheuristics

- (1) Use the modified metaheuristics algorithm to optimize the weights of the individual models.
- (2) Calculate the final accuracy of the modified MM-WMVEDL model.

Step 5: Output

- (1) Classify incoming traffic as either attack or normal traffic depends on the result of the modified MM-WMVEDL model.

Experimental results and discussion

The work in this section describes the experimental setup, datasets, and evaluation metrics used to assess the functionality of the proposed MM-WMVEDL model. The paper also discusses the outcome of the evaluation of proposed work and the performance are compared with other state-of-the-art intrusion detection methods on internet of things (Table 1).

Experimental setup

The proposed MM-WMVEDL model is instigated using Python 3.8.5 and trained the model with the system configuration of Intel Core i7–10700 K CPU and 32 GB RAM. TensorFlow 2.4.1 library is utilized for the implementation of deep learning model. The feature selection and classification processes were performed using the Scikit-learn library.

Datasets

To assess the quality of the proposed MM-WMVEDL model, two datasets were utilized in this work: the IoT-23 dataset and the UNSW-NB15 dataset.

The IoT-23 dataset is a publicly available dataset consisting of network traffic data collected from an IoT network. The dataset contains 20,000 samples of normal traffic and 9500 samples of nine different kinds of attacks such as denial-of-service (DoS), cross-site scripting (XSS), SYN flood and SQL injection, among others.

Another publicly accessible dataset that contains network traffic data obtained from an actual network is the UNSW-NB15 dataset. The dataset contains 175,341 samples of normal traffic and 88 different types of attacks. The attacks include DoS, port scan, and brute force, among others.

Evaluation metrics

The superiority of the proposed MM-WMVEDL model was assessed using different evaluation metrics such as precision, accuracy, recall, F1-score and area under the receiver operating characteristic curve (AUC-ROC).

Discussion

The proposed MM-WMVEDL model was evaluated on both the IoT-23 and UNSW-NB15 datasets. The comparative analysis with other state-of-the-art intrusion detection systems for IoT is done to investigate the functionality of the work proposed here. This analysis determines the effectiveness and superiority of the MM-WMVEDL model in detecting and mitigating intrusions within IoT networks. The tables provide detailed insights into various performance metrics such as accuracy, AUC-ROC, precision, recall, and F1-score. These metrics serve as benchmarks for evaluating the performance of the MM-WMVEDL model in relation to its counterparts. Table 2 presents the comparison of the proposed MM-WMVEDL model with other systems on the IoT-23 dataset, while Table 3 displays the comparison on the UNSW-NB15 dataset.

Fig. 2 shows the accuracy analysis of proposed work using IoT-23 dataset. Table 2 shows that the proposed MM-WMVEDL scheme attained the highest accuracy of 98.12 % on the IoT-23 dataset, outperforming all other state-of-the-art intrusion detection systems. The work also accomplished a high AUC-ROC of 0.9982, indicating its ability to discriminate between normal and attack traffic. The precision, recall, and F1-score of the MM-WMVEDL were also higher than those of other state-of-the-art intrusion detection systems.

Fig. 3 shows the accuracy of proposed work with current methods. Table 3 shows that the proposed MM-WMVEDL model increase the accuracy with the rate of 97.34 % on the UNSW-NB15 dataset, which is higher than most of the other currently available intrusion detection systems. The work also attained a high AUC-ROC of 0.9994, indicating its ability to discriminate between normal and attack traffic. The precision, recall, and F1-score of the proposed work were found to be higher than those of other state-of-the-art intrusion detection systems. The confusion matrix of the proposed work is provided in Table 4.

Conclusion

This paper presents an efficient deep learning model called MM-WMVEDL for efficient detection of IDS in IoT environment. The proposed model utilizes a multi-modal architecture to effectively capture the complex relationships between different types of network traffic data. The model also incorporates a wavelet-based feature extraction method to enhance the discriminative power of the extracted features.

The proposed work was evaluated on three publicly available datasets: IoT-23, UNSW-NB15, and CICIDS2017. Its performance was compared to several state-of-the-art intrusion detection systems. The outcomes of the work confirmed that the proposed model surpassed all the current schemes in terms of accuracy, AUC-ROC, precision, recall, and F1-score.

The proposed MM-WMVEDL model has several advantages over traditional intrusion detection systems. Firstly, it can effectively handle the multi-modal nature of network traffic data, which is common in IoT networks. Secondly, it utilizes wavelet-based feature extraction, which is known to be effective in capturing both local and global patterns in data. Thirdly, the model is highly accurate and can effectively detect both known and unknown attacks in IoT networks.

In future, the focus will be on improving the proposed model by incorporating other types of data such as network topology and device metadata. The model can also be extended to handle dynamic and evolving IoT networks.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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